

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. (All Branches) Examination
May 2011

CL101 Fundamentals of Civil Engineering

Date: 17.05.2011, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Enlist various branches of Civil Engineering & Enlist main roles or duties of a civil engineer. 07

- Q - 2 (a) Define surveying & levelling. Enlist uses of surveying. 05
(b) Differentiate between plan and map. 02
(c) Explain methods of chaining on sloping ground. 07

OR

- Q - 2 (a) Explain fundamental principles of surveying with neat sketch. 05
(b) Differentiate between plane & geodetic surveying. 02
(c) What are cumulative and compensating errors? Enlist sources of errors in chaining. 07
Explain errors due to incorrect chain.

- Q - 3 (a) Differentiate between prismatic & surveyor compass. 03
(b) The fore bearing and back bearing of a closed traverse ABCD is as under. Calculate the interior angles of the traverse. 05

Line	FB	BB
AB	$45^{\circ} 00'$	$225^{\circ} 00'$
BC	$123^{\circ} 30'$	$303^{\circ} 30'$
CD	$181^{\circ} 00'$	$1^{\circ} 00'$
DA	$289^{\circ} 00'$	$109^{\circ} 00'$

- (c) State & explain the components of GIS. 06

OR

- Q - 3 (a) Convert WCB 49° , 240° , 133° into reduced bearing. 03
(b) Explain magnetic declination. The magnetic bearing of a line AB is $S 28^{\circ} 30' E$. Calculate the true bearing if the declination is $7^{\circ} 30' W$. 05
(c) The following consecutive readings were taken with a dumpy level along a chain line at a common interval of 5m. The position of the instrument was changed after 3rd, 7th and 10th readings. Draw out the form of a level book and enter the above readings properly. Assume the RL of the 1st point as 50.000 m. Calculate RL of all points and apply the usual checks. Use any method. 06
0.350, 0.580, 0.645, 1.850, 1.935, 2.450, 1.750, 0.360, 0.720, 1.200, 0.920 and 0.630

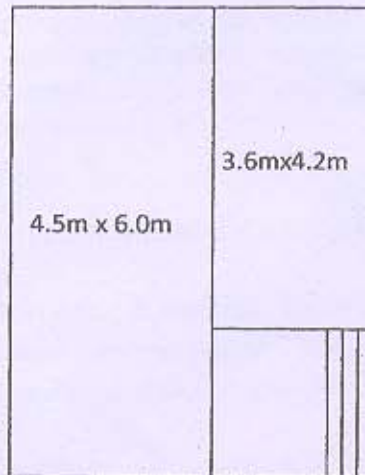
SECTION - II

Q - 4 Enlist advantages & disadvantages of roadways and railways. **07**

OR

Q - 4 Write short note on BOT projects for highways. **07**

Q - 5 (a) Prepare detailed plan of a residential building with appropriate scale from the line diagram given below. Also locate the positions of doors and windows in the plan. **06**



Q - 5 (b) Discuss various loads acting on a building in detail. **06**

Q - 5 (c) Enlist various principles of planning and explain any two in detail. **06**

OR

Q - 5 (c) Discuss in detail the classification of the buildings based on structure. **06**

Q - 6 **Attempt any two.** **10**

- (a) Explain the necessity of conservation and development of water resources in a region.
- (b) Explain with neat sketch hydrological cycle. Enlist applications of hydrology.
- (c) Describe water conveyance system.
